

Population

The totality of all items in any field of statistical investigation constitute a population.

The population can be finite or infinite. Usually size of population is denoted by 'N'.

Sample

Sample is the part of the population which is selected according to certain rules so as to obtain a value inference. The size of the sample is usually denoted by 'n'.

Remark:

If $n < 30$, we call it a small sample, otherwise large sample.

Standard Error:

The standard deviation of the sampling distribution is called standard error.

Sample - statistics	$\bar{x}, s^2$	$\hat{p}$
population - parameters	μ, σ^2	p

Testing of Hypothesis

Case I: Hypothesis of single mean

Null Hypothesis:-

The hypothesis which we want to disprove. It is usually denoted by H_0 .

Alternate Hypothesis:-

The negation of the Null Hypothesis. It is usually denoted by H_1 or H_a .

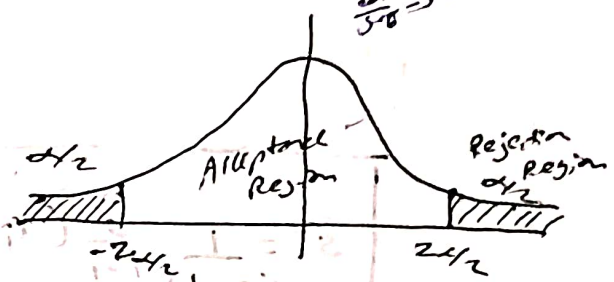
If Null hypothesis is defined as $H_0: \mu = \mu_0$, then the alternate hypothesis H_1 may be any one of the following three:

$H_1: \mu \neq \mu_0$ (Two-tailed)
or $H_1: \mu < \mu_0$ (Left-tailed)

or $H_1: \mu > \mu_0$ (Right-tailed)

Z values for two-tailed

α	1%	5%	10%
$Z_{\alpha/2}$	2.58	1.96	1.65



Level of significance

The probability of rejecting H_0 when it is true is called level of significance.

5% level of significance means there is a risk of 5% of rejecting H_0 .

Types of errors :-

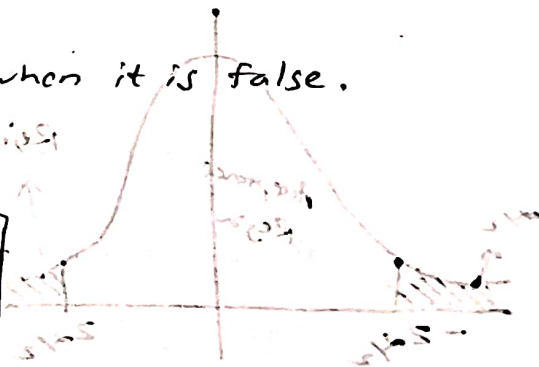
i. Type I error -

The error of rejecting H_0 when it is true.

ii. Type II error -

The error of accepting H_0 when it is false.

	Accept H_0	Reject H_0
It is true	Correct	Type I error
It is false	Type II error	Correct



Procedure for testing of hypothesis

1. Formulate the null hypothesis H_0 such that $\mu = \mu_0$. Then, the

alternative hypothesis will be any one of the three forms

$$H_1: \mu \neq \mu_0 \quad \text{(Two-tailed)}$$

$$H_1: \mu < \mu_0$$

$$H_1: \mu > \mu_0$$

2. Specify the level of significance. Normally, $\alpha = 1\%$ or 5% or 10%

3. Select an appropriate test statistic

$$Z = \frac{\bar{X} - \mu}{SE}$$

, where $SE = \frac{\sigma}{\sqrt{n}}$ (if σ is known)

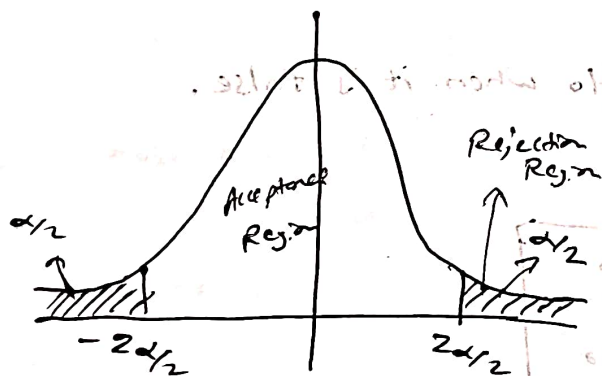
or $SE = \frac{S}{\sqrt{n}}$ (if σ is unknown)

$$S^2 = \frac{1}{(n-1)} \sum_{i=1}^n (x_i - \bar{x})^2$$

4. If $n < 30$, is small sample, we use Student's - t distribution table. If $n \geq 30$, is large sample, we use standard normal table.

5. ~~For a two tail, ie, $(H_1: \mu \neq \mu_0)$~~

Case I: For a two tail, ie, $(H_1: \mu \neq \mu_0)$



~~Case I: For a two tail, ie, $(H_1: \mu \neq \mu_0)$~~

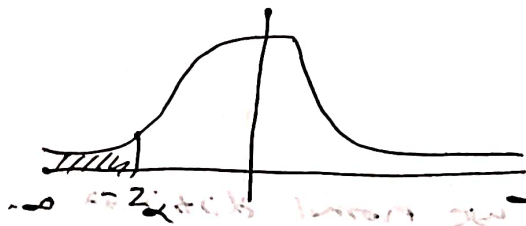
(i) if $-Z_{\alpha/2} < Z < Z_{\alpha/2} \Rightarrow |Z| < Z_{\alpha/2}$

we accept $H_0 \rightarrow$ Reject H_1

(ii) if $Z < -Z_{\alpha/2} \sim Z > Z_{\alpha/2} \Rightarrow |Z| > Z_{\alpha/2}$

we reject $H_0 \rightarrow$ Accept H_1

Case II: For a left tailed test ($H_1: \mu < \mu_0$)



(i) if $Z > -Z_\alpha$,

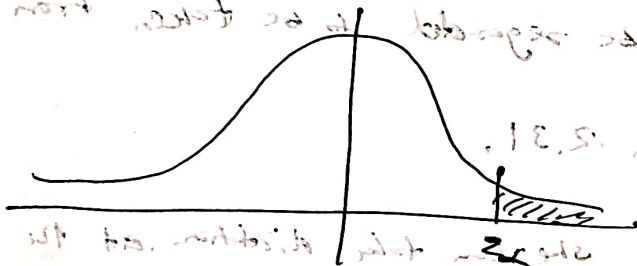
we accept $H_0 \rightarrow$ ~~Accept~~ ^{Reject} H_1

(ii) if $Z < -Z_\alpha$

we Reject $H_0 \rightarrow$ Accept H_1

$$\frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Case III: For a right tailed test ($H_1: \mu > \mu_0$)



(i) if $Z < Z_\alpha$,

we Accept $H_0 \rightarrow$ Reject H_1

(ii) if $Z > Z_\alpha$,

we Reject $H_0 \rightarrow$ Accept H_1

Q. A sample of 900 pens is found to have a mean 3.47 cm. Can it be reasonably regarded as a simple sample from a large population with mean 3.23 cm and std 2.31 cm?

$$n = 900$$

$$\bar{x} = 3.47$$

$$\mu = 3.23$$

$$\sigma = 2.31$$

$$H_0: \mu = 3.23$$

$$H_1: \mu \neq 3.23$$

$$\text{Let } \alpha = 5\%$$

$\therefore n = 900$ (large sample), we use normal distribution

$$\text{when } \alpha = 5\%, Z_{\alpha/2} = 1.96$$

$$Z = \frac{\bar{x} - \mu}{SE} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = -3.1168$$

$$\text{Here } |Z| > Z_{\alpha/2}$$

So we reject H_0

$\therefore$ The sample cannot be regarded to be taken from the population with mean 3.23 and S.D. 2.31.

Q. A stenographer claims that she can take dictation at the rate of 120 wpm. Can we reject her claim on the basis of 100 trials in which she demonstrates a mean of 116 words and S.D. of 15 words? Use 5% level of significance.

$$n = 100$$

$$\mu = 120$$

$$\bar{x} = 116$$

$$S = 15$$

$$H_0: \mu = 120$$

$$H_1: \mu \neq 120$$

Given $\alpha = 5\%$ and $n = 100$ (large sample)

$\therefore$ So we use normal distribution

$$\therefore Z_{\alpha/2} = 1.96 \text{ (from std. normal table)}$$

$$Z = \frac{\bar{X} - \mu}{SE} = \frac{\bar{X} - \mu}{S/\sqrt{n}} = -2.667$$

Here $|Z| > Z_{\alpha/2}$, So we reject H_0 .

So the stereo's represent claim is rejected.

Q). A machine is expected to produce nails of length 2 cm. A random sample of 25 nails gave an average length 2.1 cm and sd 0.25 cm. Can it be said that the machine is producing nail as per specification.

$$\mu = 2 \text{ cm}$$

$$n = 25$$

$$\bar{X} = 2.1$$

$$S = 0.25$$

$$H_0 : \mu = 2$$

$$H_1 : \mu \neq 2$$

Let ~~$\alpha = 5\%$~~ ^{here} $n = 25$ (small sample)

So we use student's -t distribution

Let ~~$\alpha = 5\%$~~

$$\therefore \alpha/2 = 2.5\% = 0.025$$

$$\text{Degrees of freedom} = n - 1$$

$$\therefore \text{Hence d.f} = 24.$$

$$\text{At df } 24, Z_{\alpha/2} = Z_{0.025} = 2.064$$

$$Z = \frac{\bar{X} - \mu}{SE} = \frac{\bar{X} - \mu}{S/\sqrt{n}} = 2$$

$(\bar{X} - \mu)$	$\bar{X} - \mu$	df
2.5	2.0	92
1	2.0	52
1	2.0	52
1	2.0	52
0	0.0	22
1	1.0	22
1	5.0	52
1	2.0	22
1	2.0	22
0	0.0	12
0.5		0.22

$$Z = \frac{\bar{X} - \mu}{SE} = 2$$

$$S(\bar{X} - \mu) = \frac{1}{\sqrt{n}}$$

$$Z = \frac{\bar{X} - \mu}{SE} = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

$$P.O.E = 2.064$$

Here $|Z| < Z_{\alpha/2}$

$\therefore$ we accept H_0 .

Q) 10 students are selected at random from a school and their heights are found to be in inches: 50, 52, 52, 53, 55, 56, 57, 58, 58, 59. In the light of these data, discuss the suggestion that the mean height of students in the school is 54 inches. Use 5% level of significance.

$n = 10$

$\mu = 54$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
50	-5	25
52	-3	9
52	-3	9
53	-2	4
55	0	0
56	1	1
57	2	4
58	3	9
58	3	9
59	4	16
550		86

$\bar{x} = \frac{\sum x_i}{n} = 55$

$$s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$$

$$= \frac{86}{9} = 9.55$$

$\therefore s = \sqrt{9.55} = 3.09$

At $\alpha = 5\%$, Here $n = 10$, $\therefore$ we use Student's t -distribution.

At $df = 9$, $Z_{\alpha/2} = Z_{0.025} = 2.262$

Test statistic, $Z = \frac{\bar{X} - \mu}{SE} = \frac{\bar{X} - \mu}{s/\sqrt{n}}$

$$Z = \frac{\bar{X} - \mu}{SE} = \frac{\bar{X} - \mu}{s/\sqrt{n}} = \frac{55 - 54}{3.01/\sqrt{10}} = 1.023$$

Since $|Z| < Z_{\alpha/2}$, we accept H_0 .

Here, $|Z| < Z_{\alpha/2} \therefore$ Accept H_0 .

Hypothesis concerning difference of means of two samples

Let $\bar{X}_1$ and $\bar{X}_2$ be the means of two samples. Then the test sample is defined as

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{SE} \quad \text{--- (1)}$$

Case 1: If sample is large, (1) $\Rightarrow$ Std. Normal distribution

Case 2: If sample is small, (1) $\Rightarrow$ Student's t distribution, with

$$d.f. \quad n_1 + n_2 - 2 \quad [(n_1 - 1) + (n_2 - 1)]$$

If, $\sigma_1^2, \sigma_2^2 \rightarrow$ population variance

$S_1^2, S_2^2 \rightarrow$ sample variance

$n_1, n_2 \rightarrow$ sample sizes

$$\frac{1}{n_1} + \frac{1}{n_2} \sqrt{\sigma_1^2 + \sigma_2^2} = SE$$

$$\frac{1}{n_1} + \frac{1}{n_2} \sqrt{S_1^2 + S_2^2} = SE$$

(i) $SE = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$, when population variance are known.

(ii) $SE = \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$, when population variance are unknown

(iii) If both samples are taken from same population (homogeneous population), then,

$$SE = \sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$Z = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Q) The means of two random samples of sizes 1000 and 2000 are 67.5 and 68 inches, respectively. Can the samples be regarded to have been drawn from the same population of std. deviation 9.5 inches. (Test at 5% level of significance).

$$n_1 = 1000$$

$$n_2 = 2000$$

$$\bar{X}_1 = 67.5$$

$$\bar{X}_2 = 68$$

$$\sigma = 9.5$$

$$H_0: \mu_1 = \mu_2 \quad \text{[null hypothesis]}$$

$$H_1: \mu_1 \neq \mu_2$$

$$SE = \sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$= 9.5 \sqrt{\frac{1}{1000} + \frac{1}{2000}}$$

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{SE}$$

$$= \frac{67.5 - 68}{9.5 \sqrt{\frac{1}{1000} + \frac{1}{2000}}}$$

$$= -1.3589$$

At $\alpha = 5\%$, $Z_{\alpha/2} = 1.96$

i.e., $|Z| < Z_{\alpha/2}$

$\therefore$ Accept H_0 .

So both samples are taken from same population.

Q) Given the following data of two distributions.

	Mean	SD	Sample Size
A	100	12	80
B	95	10	70

Test whether the difference between sample mean is significant.

$H_0: \mu_1 = \mu_2$

$H_1: \mu_1 \neq \mu_2$

$\bar{X}_1 = 100$

$\bar{X}_2 = 95$

$S_1 = 12$

$S_2 = 10$

$n_1 = 80$

$n_2 = 70$

Test statistic, $Z = \frac{\bar{X}_1 - \bar{X}_2}{SE}$

Here, $SE = \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$

$\therefore Z = \frac{100 - 95}{\sqrt{\frac{12^2}{80} + \frac{10^2}{70}}}$

$= 2.78$

Let $\alpha = 5\%$.

$\therefore Z_{\alpha/2} = 1.96$

i.e., $|Z| > Z_{\alpha/2}$

∴ Reject H_0 .

So the difference between the sample means is significant.

Q). A certain stimulus administered to each of 12 patients resulted in the following changes in blood pressure:-

5, 2, 8, -1, 3, 0, -2, 1, 5, 0, 4, 6

Can it be concluded that the stimulus will ~~in general~~ be accompanied by an increase in blood pressure?

$H_0: \mu_1 = \mu_2$

$H_1: \mu_1 < \mu_2$ [left tailed (one tailed)]

Let, difference in blood pressure $d = x_1 - x_2$

$$\bar{X} = \frac{\sum d}{n} = \frac{31}{12} = 2.583$$

$$s^2 = \frac{1}{n-1} \sum (d_i - \bar{X})^2 = 9.539$$

$$s = 3.088$$

$$SE = \frac{s}{\sqrt{n}} = \frac{3.088}{\sqrt{12}}$$

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{SE} = \frac{2.583}{\frac{3.088}{\sqrt{12}}}$$

d	d^2	$d - \bar{X}$	$(d - \bar{X})^2$
5	25		
2	4		
8	64		
-1	1		
3	9		
0	0		
-2	4		
1	1		
5	25		
0	0		
4	16		
6	36		
31	185		

$$CP = 0.001 = 5$$

$$37.5 =$$

$$CP = 0.001$$

$$37.5 < 151$$

$$Z = 2.897$$

Sample size: 12 (small sample)

$\therefore$ use student's t-distribution

$$\text{At } \alpha = 5\% \text{ or } \frac{5}{100} = 0.05$$

$$Z_{0.05} \text{ at } n-1 = 11 \text{ is } 2.201$$

$$\therefore |Z| > Z_{\alpha}$$

$\therefore$ Reject H_0

Hypothesis concerning single proportion

Null hypothesis, $H_0: p = p_0$

Alternate hypothesis, $H_1: p \neq p_0$

or $H_1: p > p_0$

or $H_1: p < p_0$

Test statistic,

$$Z = \frac{\hat{p} - p}{SE}$$

$$SE = \sqrt{\frac{pq}{n}}$$

$$q = 1 - p$$

where $\hat{p}$ is sample proportion
 p is population proportion

Q) A coin is tossed 10000 times and head turns up 5195 times.
Is the coin unbiased?

$$H_0: p = \frac{1}{2}$$

$$H_1: p \neq \frac{1}{2}$$

$$\hat{p} = \frac{5195}{10000} = 0.5195$$

$$\hat{q} = 1 - \hat{p} = 0.4805$$

test stat is,

$$Z = \frac{\hat{p} - p}{SE} = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}} = \frac{0.5195 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{10000}}} = 3.9$$

$$\text{At } \alpha = 5\%, \quad Z_{\alpha/2} = 1.96$$

$$\therefore |Z| \geq Z_{\alpha/2}$$

So, reject H_0 ,

Hence, the coin is biased.

Q) In a big city, 325 men out of 600 men were found to be smokers. Does this information support the conclusion that majority of men in the city are smokers, [use 5% level of significance].

$$\hat{p} = \frac{325}{600}$$

$$H_0 : p = 1/2$$

$$H_1 : p > 1/2 \quad (\text{majority}) \quad [\text{Right tailed}]$$

$$n = 600$$

$$p = 1/2, \quad q = 1/2$$

$$Z = \frac{\hat{p} - p}{SE} = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}} = \frac{0.5416 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{600}}} = 2.091$$

$$\text{At } \alpha = 5\%, \quad Z_{\alpha} = 1.65 \quad (\because \text{right tailed})$$

$$\therefore |Z| > Z_{\alpha}$$

$\therefore$ Reject H_0 .

$\therefore$ majority of men are smokers.

Hypothesis concerning difference of proportions

$$H_0 : p_1 = p_2, \text{ Null hypothesis}$$

$$H_1 : p_1 \neq p_2$$

$$\text{or } H_1 : p_1 > p_2$$

$$\text{or } H_1 : p_1 < p_2$$

Alternate Hypothesis

$$\text{Test statistic, } Z = \frac{\hat{p}_1 - \hat{p}_2}{SE}$$

$$(i) \text{ where } SE = \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}$$

if both samples are taken from same population (homogeneous population)

$$\text{then, } SE = \sqrt{p_0 q_0 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

$$\text{where } p_0 = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{n_1 + n_2}$$

Q) In a random sample of 1000 persons from town A, 400 are found to be consumers of wheat. In a sample of 800 from Town B, 400 were found to be consumers of wheat. Do these data reveal a significant difference between town A and town B so far as the proportion of wheat consumers is concerned?

$$H_0 : p_1 = p_2$$

$$H_1 : p_1 \neq p_2$$

$$\hat{p}_1 = \frac{400}{1000} = 0.4$$

$$\hat{p}_2 = \frac{400}{800} = 0.5$$

$$n_1 = 1000$$

$$n_2 = 800$$

Since the two samples are taken from same population (wheat consumers)

$$SE = \sqrt{p_0 q_0 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

$$\text{where, } p_0 = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{n_1 + n_2}$$

$$= 0.45$$

$$\hat{q}_0 = 1 - p_0 = 0.55$$

$$\therefore SE = \sqrt{0.45 \times 0.55 \left(\frac{1}{1000} + \frac{1}{800} \right)}$$

$$= 0.0235$$

$$\text{Test statistic, } Z = \frac{\hat{p}_1 - \hat{p}_2}{SE}$$

$$= \frac{0.4 - 0.5}{0.0235}$$

$$= -4.25$$

$$\text{Here } |Z| > Z_{\alpha/2}$$

$\therefore$ Reject H_0

$\therefore$ So there is a significant difference in the wheat consumption of two states.

Q) A machine puts out 16 imperfect articles in a sample of 500. After machine is overhauled, it puts out 3 imperfect article in a batch of 100. Has the machine improved? Test at 5% level of significance.

$$n_1 = 500$$

$$n_2 = 100$$

$$Z_{0.05} = \frac{0.05}{0.05} = 1$$

$$0.05 = 1$$

$$\hat{p}_1 = \frac{16}{500} = 0.032$$

$$\hat{p}_2 = \frac{3}{100} = 0.03$$

$$H_0 : p_1 = p_2$$

$$H_1 : p_1 > p_2 \quad [\text{non-tailed}]$$

~~$$SE = \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}$$~~

$$SE = \sqrt{p_0 q_0 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} \quad \because \text{homogeneous population.}$$

$$p_0 = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{n_1 + n_2} = 0.0317$$

$$q_0 = 0.9683$$

$$\therefore SE = \sqrt{0.0317 \times 0.9683 \left(\frac{1}{500} + \frac{1}{100} \right)}$$

$$\begin{aligned} \therefore Z &= \frac{\hat{p}_1 - \hat{p}_2}{SE} \\ &= \frac{0.032 - 0.03}{0.0192} \\ &= \underline{\underline{0.1042}} \end{aligned}$$

$$\text{At } \alpha = 5\%, \quad Z_\alpha = 1.65$$

$$\text{Here } |Z| < Z_\alpha$$

$\therefore$ Accept H_0

So the machine has not improved.